

Hrsh Venket

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Summary

ML Engineer with proven experience in developing scalable pipelines, training deep learning models using Python frameworks, and optimizing algorithms for real-world challenges. Demonstrated expertise through research contributions and hands-on work in large-scale data analysis. Seeking to leverage strong ML foundations and innovative problem-solving skills to contribute effectively to Nebula LLM's advancement.

Education

The Hong Kong University of Science and Technology

Sep 2025 - Present

Master of Science, Artificial Intelligence

Ashoka University

Aug 2021 - Jun 2025

Postgraduate Diploma in Advanced Studies and Research, Advanced Major in Computer Science

- **Coursework:** Intro to Machine Learning, Computer Vision, Numerical Algorithms and Optimisation, Computer Graphics, Statistical Inference

APL Global School

Jan 2017 - Jan 2021

A levels, A levels

Work Experience

Mphasis AI and Applied Tech Lab at Ashoka

Aug 2023 - Jul 2025

Research Assistant

Authored first-submitted paper while refining machine learning models on satellite time series data, incorporating Python-based ML libraries to enhance Indian forest fire classification accuracy by 9% over SoTA.

- Conducted large-scale data preprocessing and feature engineering for remote sensing datasets using advanced ML techniques, aligning with large-scale project methodologies.
- Presented a working paper at NASA SARI Conference 2024 to distinguished delegates, demonstrating strong analytical and problem-solving skills in AI applications.

Tiger Analytics Pvt. Limited

Oct 2024 - Dec 2024

Data Science Intern

Designed and optimized Bayesian regression, additive, and multiplicative ML models using Python, reinforcing robust machine learning principles on large-scale data projects.

- Leveraged feature engineering, validation techniques, and numerical optimization to improve forecasting accuracy in market mix modeling, showcasing effective analytical and problem-solving abilities.

Ashoka University

Aug 2023 - Dec 2024

Teaching Assistant (2 semesters)

Assisted in CS-3410: Introduction to Machine Learning and CS-4912: Computer Vision courses, engaging 120+ students and supporting 50+ office hours, thereby reinforcing core ML and deep learning concepts.

Projects

Advanced TFT Analytics Tool

Aug 2024 - Present

- Designed and deployed a web app with autoscaling on Google Cloud Platform, backed by BigQuery for large-scale data processing.
- Currently using LLMs with MCP and LangGraph to generate human readable and actionable insights from ingested data
- Built an automated autoscaling ETL pipeline from scratch to ensure seamless, reliable integration of incoming data.

Capstone Project: Predicting Forest Fire spread in Central India

Aug 2024 - Dec 2024

- Replicated DiffusionSat (Diffusion model specific to Satellite Data) to predict forest fire spread in central India
- Applied Monte Carlo Methods and to predict forest fire spread.

3D Reconstruction of C Elegans

Jan 2024 - May 2024

- Built a Computer Vision pipeline that involved image processing, 3D reconstruction and visualisation of C. Elegans from SDCM images.

Load Forecasting Bounty - Machine Learning, Electricity Load Forecasting

Jan 2024 - May 2024

- Applied gradient boosted trees and time series encoding techniques for large-scale electricity load forecasting.
- Achieved 24th/195 teams (Team: Hrsh Venket on public leaderboard)

Technical Skills

- **Programming & Domains:** Python, C/C++, PyTorch, TensorFlow, OpenCV, Computer Vision, Machine Learning, Deep Learning, Remote Sensing, Time Series Analysis, Numerical Optimization, Image Processing, Algorithms, Data Structures, Statistics, Natural Language Processing, LangGraph, RESTful APIs, Numpy, Scipy, Matplotlib, OpenCV, SciKit
- **Platforms:** Google Cloud Platform, BigQuery, Unix/Linux, Riot Developer API, Docker, Google Earth Engine

Leadership

Ashoka University

Aug 2024 - May 2025

Captain of Men's Volleyball Team

- Rebuilt squad after half of the team graduated, Captained 18 over 9 tournaments.

Ashoka University

Jan 2023 - Jan 2024

President of Eeshto (the Board Gaming Society)

- Managed a team of 50 across 6 departments, doubled membership, organised 20+ Campus events, won award for contributions to campus life

APL Global School

Jul 2020 - Jun 2021

School Pupil Leader, Secretary General